COHOMOLOGY OF THE ZEROTH LOCAL COHOMOLOGY FUNCTOR ALONG A SPECIALIZATION-STABLE SUBSET AS A FILTERED COLIMIT

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Allowed standard prerequisites (editorial).

- The functor RH^0_T of sections with support in a specialization-stable subset T, defined as the filtered colimit of the functors H^0_Z over closed Z contained in T. Defined in the author's section on generalities about local cohomology, where the counted statement occurs. Supplies the object whose cohomology the statement computes.
- K-injective complexes as a model for the derived category, and exactness of filtered colimits. Used inside the counted proof at source lines 1006-1017. Lets the proof represent the derived functor by an explicit complex and commute the colimit with cohomology.

Lemma 0.1. *Let A be a Noetherian ring. Let $T \subset \mathrm{Spec}(A)$ be a subset stable under specialization. For any object K of $D(A)$ we have*

$$H^i(RH^0_T(K)) = \mathrm{colim}_{Z \subset T \text{ closed}} H^i_Z(K)$$

Proof. Let $J^\bullet$ be a K-injective complex representing K. By definition RH^0_T is represented by the complex

$$H^0_T(J^\bullet) = \mathrm{colim}\, H^0_Z(J^\bullet)$$

where the equality follows from our definition of H^0_T. Since filtered colimits are exact the cohomology of this complex in degree i is $\mathrm{colim}\, H^i(H^0_Z(J^\bullet)) = \mathrm{colim}\, H^i_Z(K)$ as desired. □

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